

Read the passage below, making note of any important points and definitions.

EMPIRICISM & RATIONALISM

The word “empirical” means founded upon experience and observation. Empirical observations are certainly a necessary ingredient of science, and one might even say that empirical results are the foundation upon which scientific thinking is built. But can science be built on empirical results alone? Although some people might answer yes to this question, such people are in the minority these days. A brief historical sketch reveals rising and falling fortunes for empiricism in European thought. The Greeks were rather more inclined toward metaphysical speculation than empirical observation; but there was also a strain of empiricism in Greek thought, as in the biological observations of Aristotle and in the attention paid to astronomical observations. Empiricism waned during the Middle Ages, as thinkers relied more on the authority of the ancients, but it became prominent in the sixteenth and seventeenth centuries as part of the new learning advocated by natural philosophers. Galileo, for example, effectively used new observational discoveries (Jupiter’s moons and the phases of Venus) to argue for the Copernican system. In England, Francis Bacon enshrined empirical observation as the basis for all progress. Bacon is a sort of patron saint of empiricism; he envisioned the workings of science as primarily a systematic gathering of empirical facts. These facts would then be sifted and organized so as to arrive at general conclusions. Bacon’s ideas remained very influential for many years, but the actual practice of science developed a more theoretical strain in parallel with such empiricism. Currently, most people would surely say that empirical observations and theoretical constructions must mutually reinforce each other in order for science to progress.

There is an important problem with basing scientific conclusions solely on empirical facts, namely, the so-called problem of induction. Induction, or inductive logic, basi-

cally means that we can conclude that something will always happen because it has always happened before. (“I know the sun will rise tomorrow, because it has risen every day before until now.”) This logic is at the core of any attempt to prove a conclusion by empirical observation, because any such set of observations must be finite in number and yet the conclusion is meant to be general. To put it more grandiosely, we wish to make a universal statement based on an isolated set of observations. The problem with this reasoning is that there is no way we can guarantee that it is true. The fact that something has always happened before does not insure that it will always happen.

At the opposite pole of thought is the doctrine of rationalism. The champion of rationalism was René Descartes, who distrusted empirical information as a secure foundation for knowledge because the senses can be fooled. We all know that we can be seriously misled by optical illusions and so on. If the evidence of our senses can’t be trusted, what can? Descartes believed that deductive logic, exemplified by mathematics, was the surest foundation for knowledge. Ultimately, then, science should be based on our minds, not our senses. Rationalism, taken to an extreme, proclaims that truth can be apprehended directly by our minds. We only need to use the proper thought processes, identifying manifestly true first principles, and deducing their consequences; no empirical input is needed in this view. Few people still accept this extreme view. Historically, neither empiricism nor rationalism has proven adequate by itself as a philosophical foundation for science. Perhaps the most important result of Descartes’ program was to highlight the value of mathematics in the sciences.

Gregory N Derry. *What science is and how it works*. Princeton University Press, 1999.